

Automata on Infinite Trees

Büchi Automata on Infinite Trees

Definition

A Büchi tree automaton over Σ is $A = \langle S, I, T, F \rangle$, where:

- S is a finite set of *states*,
- $I \subseteq S$ is a set of *initial states*,
- $T : S \times \Sigma \rightarrow 2^{S \times S}$ is the *transition relation*,
- $F \subseteq S$ is the set of *final states*.

Runs

A *run* of A over a tree $t : \{0, 1\}^* \rightarrow \Sigma$ is a mapping $\pi : \{0, 1\}^* \rightarrow S$ such that, for each position $p \in \{0, 1\}^*$, where $q = \pi(p)$, we have:

- if $p = \epsilon$ then $q \in I$, and
- if $q_i = \pi(pi)$, $i = 0, 1$ then $\langle q_0, q_1 \rangle \in T(q, t(p))$.

If π is a *run* of A and σ is a *path* in t , let $\pi|_{\sigma}$ denote the path in π corresponding to σ .

A run π is said to be *accepting*, if and only if for every path σ in t we have:

$$\text{inf}(\pi|_{\sigma}) \cap F \neq \emptyset$$

Closure Properties

For every Büchi automaton A there exists a complete Büchi automaton A' such that $\mathcal{L}(A) = \mathcal{L}(A')$.

Theorem 1 *The class of Büchi-recognizable tree languages is closed under union, intersection and projection.*

Let $A_i = \langle S_i, I_i, T_i, F_i \rangle$, $i = 1, 2$, where $S_1 \cap S_2 = \emptyset$.

Let $A_{\cup} = \langle S_1 \cup S_2, I_1 \cup I_2, T_1 \cup T_2, F_1 \cup F_2 \rangle$.

Closure Properties

Let $A_{\cap} = \langle S, I, T, F \rangle$ where:

- $S = S_1 \times S_2 \times \{0, 1, 2\}$
- $I = I_1 \times I_2 \times \{1\}$
- for any $s, s_1, s_2 \in S_1, s', s'_1, s'_2 \in S_2, a, b \in \{0, 1, 2\}$:

$$\langle (s_1, s'_1, b), (s_2, s'_2, b) \rangle \in T((s, s', a), \sigma)$$

iff $\langle s_1, s_2 \rangle \in T(s, \sigma), \langle s'_1, s'_2 \rangle \in T(s', \sigma)$ and:

1. if $a = 0$ or $(a = 1 \text{ and } s \notin F_1)$, then $b = 1$
 2. if $(a = 1 \text{ and } s \in F_1)$ or $(a = 2 \text{ and } s' \notin F_2)$, then $b = 2$
 3. if $a = 2$ and $s' \in F_2$, then $b = 0$
- $F = S \times S \times \{0\}$

Emptiness of Büchi Tree Automata

Let $A = \langle S, I, T, F \rangle$ be a Büchi tree automaton where $F = \{s_1, \dots, s_m\}$, and $\pi : \{0, 1\}^* \rightarrow S$ be an **accepting run** of A on the tree $t \in \mathcal{T}(\Sigma)$.

For any $s \in S$, and any $u \in \{0, 1\}^*$ such that $\pi(u) = s$, let

$$d_u^\pi = \{w \in u \cdot \{0, 1\}^* \mid \forall v . u < v < w \Rightarrow \pi(v) \notin F\}$$

By König's lemma, d_u^π is finite for any $u \in \{0, 1\}^*$.

Let t_s^π be the restriction of t to d_u^π . Let

$$T_s = \{t_s^\pi \mid \pi \text{ is an accepting run of } A \text{ on } t\}$$

Emptiness of Büchi Tree Automata

If $\vec{s} = \langle s_1, \dots, s_m \rangle$ are the final states of A :

$$\mathcal{L}(A) = \bigcup_{s_0 \in I} T_{s_0} \cdot_{\vec{s}} \langle T_{s_1}, \dots, T_{s_m} \rangle^{\omega \vec{s}}$$

Conversely, the expression above denotes a Büchi-recognizable tree language.

Let $A = \langle S, I, T, F \rangle$ be a Büchi tree automaton. For each $s \in S$ let T_s be the recognizable tree language defined above. Eliminate from S (and T) all states s such that $T_s = \emptyset$, and let S' be the resulting set of states.

We claim that $\mathcal{L}(A) \neq \emptyset \iff S' \cap I \neq \emptyset$.

The Complement Problem

Let $\Sigma = \{a, b\}$, $\mathcal{T}_0 = \{t \in \mathcal{T}^\omega(\Sigma) \mid \text{some path in } t \text{ has infinitely many } a\text{'s}\}$

$\mathcal{T}_0$ is Büchi recognizable.

Let $A = \langle \{s_0, s_1, s_a, s_b\}, \{s_0\}, T, \{s_1, s_a\} \rangle$, where T is defined by:

$$a(s_{0,a,b}) \rightarrow \{\langle s_1, s_a \rangle, \langle s_a, s_1 \rangle\}$$

$$b(s_{0,a,b}) \rightarrow \{\langle s_1, s_b \rangle, \langle s_b, s_1 \rangle\}$$

$$a(s_1) \rightarrow \{\langle s_1, s_1 \rangle\}$$

$$b(s_1) \rightarrow \{\langle s_1, s_1 \rangle\}$$

The Complement Problem

Let $\mathcal{T}_1 = \mathcal{T}^\omega(\Sigma) \setminus \mathcal{T}_0 = \{t \in \mathcal{T}^\omega(\Sigma) \mid \text{all paths in } t \text{ have finitely many } a\text{'s}\}$.

We show that $\mathcal{T}_1$ cannot be recognized by a Büchi tree automaton.

Exercise 1 $I = \{s_0, s_1\}$, $F = \{s_1\}$ and

$$a(s_0) \rightarrow \langle s_0, s_0 \rangle$$

$$\langle s_0, s_1 \rangle$$

$$\langle s_1, s_0 \rangle$$

$$\langle s_1, s_1 \rangle$$

$$b(s_0) \rightarrow \langle s_0, s_0 \rangle$$

$$\langle s_0, s_1 \rangle$$

$$\langle s_1, s_0 \rangle$$

$$\langle s_1, s_1 \rangle$$

$$b(s_1) \rightarrow \langle s_1, s_1 \rangle$$

The Complement Problem

Let $T_n : \{0, 1\}^* \rightarrow \Sigma$ be the language of trees:

$$t_n(p) = \begin{cases} a & \text{if } p \in \{\epsilon, 1^{m_1}0, 1^{m_1}01^{m_2}0, \dots, 1^{m_1}01^{m_2}0 \dots 1^{m_n}0 \mid m_1, \dots, m_n \in \mathbb{N}\} \\ b & \text{otherwise} \end{cases}$$

Obviously, $T_n \subset \mathcal{T}_1$, for all $n \in \mathbb{N}$.

Suppose there exists a Büchi automaton $A = \langle S, I, T, F \rangle$ with k states, s.t. $\mathcal{L}(A) = \mathcal{T}_1$. Let π be the accepting run of A over t_{k+1} . Then there exist:

- $m_1 > 0$ such that $\pi(1^{m_1}) = s_1 \in F$
- $m_2 > 0$ such that $\pi(1^{m_1}01^{m_2}) = s_2 \in F$
- ...

There exists a path σ in t_m and $u < v < w < \sigma$, such that

$\pi(u) = \pi(w) = s \in F$ and $t_m(v) = a$. Then $\pi = r_1 \cdot_s r_2 \cdot_s r_3$, and $r_1 \cdot_s r_2^{\omega s}$ is an accepting run on $q_1 \cdot q_2^\omega$, which contains a path with infinitely many a .

Muller Automata on Infinite Trees

Definition

A **Muller** tree automaton Σ is $A = \langle S, I, T, \mathcal{F} \rangle$, where:

- S is a finite set of *states*,
- $I \subseteq S$ is a set of *initial states*,
- $T : S \times \Sigma \rightarrow 2^{S \times S}$ is the *transition function*,
- $\mathcal{F} \subseteq 2^S$, is the set of *accepting sets*.

A run π of A over t is said to be *accepting*, iff for every path σ in t :

$$\inf(\pi|_{\sigma}) \in \mathcal{F}$$

Closure Properties

The class of Muller-recognizable tree languages is closed under union and intersection.

For **union**, the proof is exactly as in the case of Büchi automata. For $A_{\cup}$, the set of accepting sets is the union of the sets $\mathcal{F}_i$, $i = 1, 2$.

For **intersection**, let $A_{\cap} = \langle S_1 \times S_2, I_1 \times I_2, T, \mathcal{F} \rangle$, where:

- $\langle (s_1, s'_1), (s_2, s'_2) \rangle \in T((s, s'), \sigma)$ iff $\langle s_1, s_2 \rangle \in T(s, \sigma)$ and $\langle s'_1, s'_2 \rangle \in T(s', \sigma)$, and
- $\mathcal{F} = \{G \in S_1 \times S_2 \mid pr_1(G) \in \mathcal{F}_1 \text{ and } pr_2(G) \in \mathcal{F}_2\}$, where:
 - $pr_1(G) = \{s \in S_1 \mid \exists s' . (s, s') \in G\}$, and
 - $pr_2(G) = \{s \in S_2 \mid \exists s' . (s', s) \in G\}$.

Rabin Automata on Infinite Trees

Definition

A **Rabin** tree automaton Σ is $A = \langle S, I, T, \Omega \rangle$, where:

- S is a finite set of *states*,
- $I \subseteq S$ is a set of *initial states*,
- $T : S \times \Sigma \rightarrow 2^{S \times S}$ is the *transition function*,
- $\Omega = \{ \langle N_1, P_1 \rangle, \dots, \langle P_n, N_n \rangle \}$ is the set of *accepting pairs*.

A run π of A over t is said to be *accepting*, if and only if for every path σ in t there exists a pair $\langle N_i, P_i \rangle \in \Omega$ such that:

$$\inf(\pi|_{\sigma}) \cap N_i = \emptyset \text{ and } \inf(\pi|_{\sigma}) \cap P_i \neq \emptyset$$

Büchi, Muller and Rabin

For every Büchi tree automaton A there exists a Muller tree automaton B , such that $\mathcal{L}(A) = \mathcal{L}(B)$, but not viceversa.

For every Muller tree automaton A there exists a Rabin tree automaton B , such that $\mathcal{L}(A) = \mathcal{L}(B)$, and viceversa.

From Büchi to Muller

For each (nondeterministic) Büchi automaton A there exists a (nondeterministic) Muller automaton B such that $\mathcal{L}(A) = \mathcal{L}(B)$

Let $A = \langle S, I, T, F \rangle$ be a Büchi automaton.

Define $B = \langle S, I, T, \{G \in 2^S \mid G \cap F \neq \emptyset\} \rangle$

Allowing Muller automata to be nondeterministic is essential here.

From Rabin to Muller

Given a Rabin automaton $A = \langle S, I, T, \Omega \rangle$, such that

$$\Omega = \{ \langle N_1, P_1 \rangle, \dots, \langle N_k, P_k \rangle \}$$

let $B = \langle S, I, T, \mathcal{F} \rangle$ be the Muller automaton, where

$$\mathcal{F} = \{ F \subseteq S \mid F \cap N_i = \emptyset \text{ and } F \cap P_i \neq \emptyset \text{ for some } 1 \leq i \leq k \}$$

From Muller to Rabin

Given a Muller automaton $A = \langle S, I, T, \mathcal{F} \rangle$, there exists a Rabin automaton B such that $\mathcal{L}(A) = \mathcal{L}(B)$

Let $\mathcal{F} = \{Q_1, \dots, Q_k\}$

Let $B = \langle S', I', T', \Omega' \rangle$ where:

- $S' = 2^{Q_1} \times \dots \times 2^{Q_k} \times S$
- $I' = \{ \langle \emptyset, \dots, \emptyset, s_0 \rangle \mid s_0 \in I \}$

From Muller to Rabin

- $T'(\langle S_1, \dots, S_k, s \rangle, a) = (\langle S'_1, \dots, S'_k, s' \rangle, \langle S''_1, \dots, S''_k, s'' \rangle)$ where:
 - $T(s, a) = (s', s'')$
 - for all $1 \leq i \leq k$:

$$S'_i = S''_i = \begin{cases} \emptyset & , \text{ if } S_i \cup \{s\} = Q_i \\ (S_i \cup \{s\}) \cap Q_i & , \text{ otherwise} \end{cases}$$

- $P_i = \{\langle S_1, \dots, S_i, \dots, S_k, s \rangle \mid S_i = Q_i\}, 1 \leq i \leq k$
- $N_i = \{\langle S_1, \dots, S_i, \dots, S_k, s \rangle \mid s \notin Q_i\}, 1 \leq i \leq k$

The Rabin Complementation Theorem

Theorem 2 (Rabin '69) *The class of Rabin-recognizable tree languages is closed under complement.*

The class of Rabin-recognizable tree languages is closed under union and intersection, because Muller-recognizable languages are.

Emptiness of Rabin Automata

Given an alphabet Σ , an infinite tree $t \in \mathcal{T}^\omega(\Sigma)$ is said to be *regular* if there are only finitely many distinct subtrees t_u of t , where $u \in \{0, 1\}^*$.

Example 1 *The infinite binary tree $f(g(f(\dots), f(\dots)), g(f(\dots), f(\dots)))$ is regular. $\square$*

Theorem 3 (Rabin '72)

1. *Any non-empty Rabin-recognizable set of trees contains a regular tree.*
2. *The emptiness problem for Rabin tree automata is decidable.*

Reduction to empty alphabet

Let $A = \langle S, I, T, \Omega \rangle$ be a Rabin tree automaton over Σ , such that $\mathcal{L}(A) \neq \emptyset$, where $\Omega = \{\langle N_1, P_1 \rangle, \dots, \langle N_n, P_n \rangle\}$.

Let $A' = \langle S \times \Sigma, I \times \Sigma, T', \Omega' \rangle$, where:

- $\langle (s_1, \sigma_1), (s_2, \sigma_2) \rangle \in T'((s, \sigma))$ iff $\langle s_1, s_2 \rangle \in T(s, \sigma)$, and $\sigma_1, \sigma_2 \in \Sigma$.
- $\Omega' = \{\langle N_1 \times \Sigma, P_1 \times \Sigma \rangle, \dots, \langle N_n \times \Sigma, P_n \times \Sigma \rangle\}$.

The accepting runs of A' are pairs (π, t) , where $t \in \mathcal{L}(A)$, and π is a accepting run of A on t .

Regular accepting runs

For any Rabin tree automaton A , there exists a Rabin tree automaton A' **with one initial state** such that $\mathcal{L}(A) = \mathcal{L}(A')$.

Consider a Rabin tree automaton $A = \langle S, s_0, T, \Omega \rangle$ over the empty alphabet, and let π be a accepting run of A .

Claim 1 *If A has a accepting run, A has also a **regular** accepting run.*

A state $s \in S$ is said to be **live** if $s \neq s_0$ and $\langle s_1, s_2 \rangle \in T(s)$ for some $s_1, s_2 \in S$, where either $s_1 \neq s$ or $s_2 \neq s$.

By induction on $n =$ the number of live states in A .

Regular accepting runs

Base case $n = 0$: $\pi(\epsilon) = s_0$ and $\pi(p) = s$, for all $p \in \text{dom}(\pi)$, and $s \in S$ is non-live.

Inductive step $n > 0$:

Case 1 If some live state in A is missing on π , apply the induction hypothesis.

Case 2 All live states of A appear on π , and there is a position $u \in \{0, 1\}^*$ such that $\pi(u) = s$ is live, but some live state s' does not appear in π_u .

Let $\pi_1 = \pi \setminus \pi_u$ and $\pi_2 = \pi_u$. Both π_1 and π_2 are runs of automata with $n - 1$ live states, hence there exists accepting regular runs π'_1 and π'_2 of these automata. The desired run is $\pi'_1 \cdot_s \pi'_2$.

Regular accepting runs

Case 3 All live states appear in any subtree of π . Let σ be a path in π consisting of all the live states appearing again and again, and only of the live states, with the exception of $\pi(\epsilon)$. **Q: Why does σ exist?**

There exists $\langle N, P \rangle \in \Omega$, such that $\inf(\sigma) \cap N = \emptyset$ and $\inf(\sigma) \cap P \neq \emptyset$.

Then N contains only non-live states.

Let $s \in \inf(\sigma) \cap P$ and u, v be the 1st and 2nd positions such that $\sigma(u) = \sigma(v) = s$.

Let $\pi_1 = \pi \setminus \pi_u$ and $\pi_2 = \pi_u \setminus \pi_v$. Both π_1 and π_2 are accepting runs of automata with $n - 1$ live states, hence there exists accepting regular runs π'_1 and π'_2 of these automata. The desired run is $\pi'_1 \cdot_s \pi'^{\omega s}_2$.

The Emptiness Problem

Let A be an input-free Rabin tree automaton with n live states.

We derive $A_{n-1}, A_{n-2}, \dots, A_0$ from A , having $n-1, n-2, \dots, 0$ live states.

If A has a accepting run, then it it has a regular run, composed of runs of $A_{n-1}, A_{n-2}, \dots, A_0$.

So it is enough to check emptiness of $A_{n-1}, A_{n-2}, \dots, A_0$.

Rabin Automata, SkS and $S\omega S$

Defining infinite paths

We say that a set of positions X is **linear** iff the following holds:

$$linear(X) \quad : \quad (\forall x, y . X(x) \wedge X(y) \rightarrow x \leq y \vee y \leq x)$$

X is a **path** iff:

$$path(X) \quad : \quad linear(X) \wedge \forall Y . linear(Y) \wedge X \subseteq Y \rightarrow X = Y$$

From Automata to Formulae

Let $A = \langle S, I, T, \Omega \rangle$ be a Rabin tree automaton, where $S = \{s_1, \dots, s_p\}$.

Let $\vec{Y} = \{Y_1, \dots, Y_p\}$ be set variables.

If X denotes a path, state i appears infinitely often in X iff:

$$inf_i(X) : \forall x . X(x) \rightarrow \exists y . x \leq y \wedge X(y) \wedge Y_i(y)$$

The formula expressing the accepting condition is:

$$\Phi_\Omega(\vec{Y}) : \forall X . path(X) \rightarrow \bigvee_{\langle N, P \rangle \in \Omega} \left(\bigwedge_{s_i \in N} \neg inf_i(X) \wedge \bigvee_{s_i \in P} inf_i(X) \right)$$

Decidability of S2S

Theorem 4 *Given an alphabet Σ , a tree language $L \subseteq \mathcal{T}^\omega(\Sigma)$ is definable in S2S iff it is recognizable.*

Corollary 1 *The SAT problem for S2S is decidable.*

Obtaining Decidability Results by Reduction

Suppose we have a logic $\mathcal{L}$ interpreted over the domain $\mathcal{D}$, such that the following problem is decidable:

for each formula φ of $\mathcal{L}$ there exists $\mathfrak{m} \in \mathcal{D}$ such that $\mathfrak{m} \models \varphi$

Then we can prove the same thing for another logic $\mathcal{L}'$ interpreted over $\mathcal{D}'$ iff there exists functions $\Delta : \mathcal{D}' \rightarrow \mathcal{D}$ and $\Lambda : \mathcal{L}' \rightarrow \mathcal{L}$ such that for all $\mathfrak{m}' \in \mathcal{D}'$ and $\varphi' \in \mathcal{L}'$ we have:

$$\mathfrak{m}' \models \varphi' \iff \Delta(\mathfrak{m}') \models \Lambda(\varphi')$$

Decidability of $S\omega S$

Every tree $t : \mathbb{N}^* \rightarrow \Sigma$ can be encoded as $t' : \{0, 1\}^* \rightarrow \Sigma$.

Let $D = \{\epsilon\} \cup \{1^{n_1+1}01^{n_2+1}0 \dots 1^{n_k+1}0 \mid k \geq 1, n_i \in \mathbb{N}, 1 \leq i \leq k\}$.

Embedding the domain of $S\omega S$ into $S2S$:

$$D(x) \quad : \quad \exists z \forall y . z \leq y \wedge x = z \quad \vee \quad \forall y . s_0(y) \leq x \rightarrow \exists y' . y = s_1(y')$$

Decidability of $S_\omega S$

If $p = 1^{n_1+1}01^{n_2+1}0 \dots 1^{n_k+1}0$, let

$$f_i(p) = p \cdot 1^{i+1}0 = 1^{n_1+1}01^{n_2+1}0 \dots 1^{n_k+1}01^{i+1}0$$

$$x \preceq_D y \quad : \quad D(x) \wedge D(y) \wedge x \preceq y$$

Define the relation $x \leq_D^\exists y$ iff $x \in D$ and $y = x \cdot 1^{n+1}0$, for some $n \in \mathbb{N}$:

$$x \leq_D^\exists y \quad : \quad \exists z . y = s_0(z) \wedge \forall z' . x \leq z \wedge z' < z \rightarrow s_1(z') \leq y$$

Define $f_0, f_1, f_2, \dots$ by induction:

$$f_0(x) = y \quad : \quad D(x) \wedge D(y) \wedge \exists z . y = s_0(z) \wedge z = s_1(x)$$

$$f_{i+1}(x) = y \quad : \quad D(x) \wedge D(y) \wedge x \leq_D^\exists y \wedge \forall z . x \leq_D^\exists z \wedge$$

$$\bigwedge_{0 \leq k \leq i} z \neq f_k(x) \rightarrow y \preceq_D z$$